

Potential of Fecal Microbiota Transplantation to Prevent Acute GVHD: Analysis from a Phase II Trial

Armin Rashidi^{1,2,3}, Maryam Ebadi⁴, Tauseef Ur Rehman¹, Heba Elhousseini¹, David Kazadi⁵, Hossam Halaweish⁶, Mohammad H. Khan⁶, Andrea Hoeschen¹, Qing Cao⁷, Xianghua Luo^{7,8}, Amanda J. Kabage⁹, Sharon Lopez⁹, Shernan G. Holtan¹, Daniel J. Weisdorf¹, Alexander Khoruts^{9,10,11}, and Christopher Staley⁶



ABSTRACT

Purpose: Intestinal microbiota disruptions early after allogeneic hematopoietic cell transplantation have been associated with increased risk for acute GVHD (aGVHD). In our recent randomized phase II trial of oral, encapsulated, third-party fecal microbiota transplantation (FMT) versus placebo, FMT at the time of neutrophil recovery was safe and ameliorated dysbiosis. Here, we evaluated in *post hoc* analysis whether donor microbiota engraftment after FMT may protect against aGVHD.

Experimental Design: We analyzed pre- and post-FMT stool samples and estimated donor microbiota engraftment (a pre-planned secondary endpoint) by determining the fraction of post-FMT microbiota formed by unique donor taxa (donor microbiota fraction; dM_f).

Results: dM_f was higher in patients who later developed grade I or no aGVHD (median 33.9%; range, 1.6%–74.3%) than those who developed grade II–IV aGVHD (median

25.3%; range, 2.2%–34.8%; $P = 0.006$). The cumulative incidence of grade II–IV aGVHD by day 180 was lower in the group with greater-than-median dM_f than the group with less-than-median dM_f [14.3% (95% confidence interval, CI, 2.1–37.5) vs. 76.9% (95% CI, 39.7–92.8), $P = 0.008$]. The only determinant of dM_f in cross-validated least absolute shrinkage and selection operator (LASSO)-regularized regression was the patient's pre-FMT microbiota diversity (Pearson correlation coefficient -0.82 , $P = 1.6 \times 10^{-9}$), indicating more potent microbiota modulation by FMT in patients with more severe dysbiosis. Microbiota network analysis revealed major rewiring including changes in the most central nodes, without emergence of keystone species, as a potential mechanism of FMT effect.

Conclusions: FMT may have protective effects against aGVHD, especially in patients with more severe microbiota disruptions.

Introduction

Recipients of allogeneic hematopoietic cell transplantation (HCT) and patients with acute myeloid leukemia (AML) receiving induction chemotherapy experience major disruptions in their intestinal microbiota, collectively known as gut dysbiosis (1, 2). Features of gut dysbiosis have been associated with increased incidence of acute GVHD (aGVHD) and its associated mortality in several studies. Dysbiotic features in this context include loss of microbiota diversity (3, 4), loss of obligate anaerobic and butyro-

genic commensal bacteria (5–8), and expansion of *Enterococcus* (9), mucolytic bacteria (10), and picobirnaviruses (11). Although the causality of some of these associations is unclear, mechanistic evidence is accumulating. Loss of immunosuppressive and gut barrier protective microbial-derived metabolites such as short-chain fatty acids (e.g., butyrate; refs. 6, 7) and indole (12), altered microbial metabolism of GVHD prophylactic medications (e.g., mycophenolate mofetil; ref. 13), and thinning of the intestinal barrier by mucolytic species (e.g., *Bacteroides thetaiotaomicron*; ref. 10) are some of the proposed pathways via which microbiota injury may result in aGVHD.

Inspired by the repeatedly observed relationship between gut dysbiosis and poor outcomes after curative-intent therapy in allogeneic HCT recipients and patients with AML, we hypothesized that fecal microbiota transplantation (FMT) from healthy third-party donors—a major microbiota restorative intervention at the microbial community level—may not only improve microbiota indices but also improve clinical outcomes. To test this hypothesis, we conducted a single-center, randomized, double-blind, placebo-controlled phase II trial in two independent cohorts (HCT, 74 patients; AML, 26 patients), where we administered oral, encapsulated, third-party FMT versus placebo at the time of neutrophil recovery. FMT was safe, feasible, and ameliorated gut dysbiosis, but the trial did not meet the primary endpoint of reducing infection rate (14).

A secondary endpoint of the trial was grade II–IV aGVHD. There was a numerically higher incidence of grade II–IV aGVHD after FMT compared with placebo (29.8% vs. 8.3%, respectively). However, this result was confounded by a large imbalance between GVHD prophylactic regimens, where a greater proportion of patients in the placebo arm received posttransplantation cyclophosphamide (PTCy; 76.0% vs. 44.9% in the FMT arm; ref. 14). The incidence of grade II–IV aGVHD

¹Division of Hematology, Oncology, and Transplantation, Department of Medicine, University of Minnesota, Minneapolis, Minnesota. ²Clinical Research Division, Fred Hutchinson Cancer Center, Seattle, Washington. ³Division of Oncology, University of Washington, Seattle, Washington. ⁴Department of Radiation Oncology, University of Washington and Fred Hutchinson Cancer Center, Seattle, Washington. ⁵Department of Medicine, University of Minnesota, Minneapolis, Minnesota. ⁶Department of Surgery, University of Minnesota, Minneapolis, Minnesota. ⁷Biostatistics Core, Masonic Cancer Center, University of Minnesota, Minneapolis, Minnesota. ⁸Division of Biostatistics, School of Public Health, University of Minnesota, Minneapolis, Minnesota. ⁹Division of Gastroenterology, Hepatology, and Nutrition, Department of Medicine, University of Minnesota, Minneapolis, Minnesota. ¹⁰Center for Immunology, University of Minnesota, Minneapolis, Minnesota. ¹¹Biotechnology Institute, University of Minnesota, St. Paul, Minnesota.

Corresponding Author: Armin Rashidi, Fred Hutchinson Cancer Center, Seattle, WA 98109. E-mail: arashidi@fredhutch.org

Clin Cancer Res 2023;29:4920–9

doi: 10.1158/1078-0432.CCR-23-2369

©2023 American Association for Cancer Research

Translational Relevance

We find evidence that greater donor microbiota engraftment after third-party fecal microbiota transplantation (FMT) is associated with a lower rate of grade II–IV acute GVHD (aGVHD) in patients undergoing allogeneic hematopoietic cell transplantation. In addition, we find pre-FMT microbiota diversity to be the only determinant of engraftment level, with lower diversity predicting greater engraftment. These findings suggest that FMT may have protective effects against aGVHD, especially in patients with more severe microbiota disruptions. Our results support a future definitive randomized trial to determine whether FMT, added to standard GVHD prophylaxis, may reduce the rates of aGVHD.

after placebo was lower than expected when compared with our institutional rates of grade II–IV aGVHD with PTCy-based GVHD prophylaxis after myeloablative HCT (~17%; ref. 15) and national rates with PTCy-based GVHD prophylaxis after reduced-intensity HCT (~50%; ref. 16). Thus, although in single-arm studies, FMT has led to clinical improvement in some patients with steroid-refractory aGVHD (17–19), whether FMT early after HCT may protect against subsequent aGVHD is unknown.

High-throughput sequencing of the microbiota in donor and pre-intervention/postintervention longitudinal stool samples from this trial allowed us to estimate donor microbiota engraftment, a pre-planned secondary endpoint. Donor microbiota engraftment is a measure of microbiota modulation by FMT and a predictor of its clinical efficacy in several disease settings (20, 21), including the treatment of steroid-refractory aGVHD (17). By analyzing donor microbiota engraftment in relation to grade II–IV aGVHD, a *post hoc* analysis of the FMT arm, we provide the first evidence connecting effective posttransplant FMT modulation of the microbiota with protection against aGVHD. In addition, the determinants of donor microbiota engraftment in the complex setting of antibiotic exposure combined with cytotoxic intestinal barrier damage and immunosuppression are unknown. We find pre-FMT microbiota diversity (a community-level property of the microbiota) rather than specific taxa, to be a key determinant of donor microbiota engraftment. Finally, we demonstrate differential effects of FMT on microbial network properties to gain insight into potential mechanisms for our findings.

Materials and Methods

Trial design summary (reported previously)

The trial protocol (ClinicalTrials.gov identifier: NCT03678493) was approved by the University of Minnesota Institutional Review Board and complied with local regulations and the Declaration of Helsinki. All patients provided written informed consent. Trial protocol, design, eligibility criteria, and procedures for this trial were published previously (14). Briefly, adults undergoing inpatient, T-replete allogeneic HCT for any indication (HCT cohort) or inpatient chemotherapy for AML (AML cohort) underwent simple, double-blinded randomization in a 2:1 ratio between two arms, namely third-party FMT or placebo. The study treatment consisted of five oral capsules taken at once after neutrophil recovery and at least 2 days after discontinuation of antibacterial antibiotics. Patients reexposed to antibacterial antibiotics were redosed, for a maximum total of three doses until 3 months after enrollment. Each patient received material manufactured from

one of the 4 donors. Each FMT capsule contained $\geq 1 \times 10^{11}$ bacteria with $\geq 40\%$ viability. Patients were followed for 9 months. Stool samples were collected in 95% ethanol at baseline (before starting conditioning; T0), pre-FMT/placebo (T1), 10 days post-FMT/placebo (T2), 28 days post-FMT/placebo (T3), and at 9 months (T4). Exact amplicon sequence variants (ASV) were inferred from the amplified V4 hypervariable region of the 16 rRNA gene using DADA2 v1.18.0 (22) and mapped using the SILVA nonredundant v138.1 training set (23). ASVs represent sequence variants differing from one another by as few as a single nucleotide. Compared with the operational taxonomic units generated by clustering sequences based on a threshold of nucleotide similarity (e.g., 97%), ASV-based methods provide a finer taxonomic resolution. Alpha diversity was measured by the Shannon index.

The primary endpoint was all-cause infection rate within 4 months after dose 1. Secondary endpoints included specific types of infection (bacterial, viral, or fungal), bloodstream infections within 7 days after each dose (safety endpoint), grade II–IV aGVHD (HCT cohort only), and donor microbiota engraftment. The trial was powered for the primary endpoint and a sample size of 72 patients in each cohort was estimated to provide 80% power to detect a 50% lower rate of infection in the FMT arm compared with the placebo arm at a two-sided alpha level of 5%.

The trial opened to accrual in September 2019 and reached its accrual goal for the HCT cohort in February 2022 after enrolling 74 HCT patients (FMT, 49; placebo, 25). Because of slow accrual in the AML cohort, this cohort was closed to new accrual at the same time, after enrolling a total of 26 patients (FMT, 18; placebo, 8). Clinical data cutoff occurred on August 5, 2022, once the last subject in each cohort completed follow-up for the last clinical endpoint. The complete clinical dataset, with no loss to follow-up or missing data, was used in the first report from this trial. Microbiome composition was similar between FMT and placebo arms at T0, and also at T1, arguing against a differential effect on the microbiome by the GVHD prophylactic regimen and permitting a fair comparison of the posttreatment microbiota between the two arms for FMT effects. FMT augmented the recovery of microbiome diversity and composition, an effect that was apparent as early as T2 and persisted through T4 (14).

Study design (current study)

We had three objectives in the current study: (i) identify the determinants of donor microbiota engraftment in post-FMT samples, (ii) determine the relationship between donor microbiota engraftment and grade II–IV aGVHD, and (iii) determine the effects of FMT on network properties of the microbiota. These objectives were addressed in *post hoc* analysis. We used all patients and samples relevant for each objective in its corresponding analysis. Specifically, and by definition, objective (i) was limited to the FMT arm of both HCT and AML cohorts, objective (ii) was limited to the FMT arm of the HCT cohort, and objective (iii) was evaluated using both arms of both cohorts. The randomized design of the trial was used in (iii) to aid in distinguishing differential effects of FMT from changes in microbiota network properties after placebo (i.e., changes unrelated to microbiota intervention including those representing spontaneous recovery from injury). T1, T2, and T3 samples were used to determine the fraction of post-FMT microbiota formed by unique donor taxa, which was then used for objectives (i) and (ii). T4 samples were not included because aGVHD events preceded their collection. T1 and T3 samples were used for objective (iii).

Statistical analysis

The analysis schema is summarized in **Fig. 1**. The analyses were limited to each subject's dose 1. In the first stage, we included all 35 patients on the FMT arm of both cohorts who had T1 and at least one of T2 or T3 samples. These were the patients for whom donor microbiota engraftment could be defined and calculated. In each post-FMT sample, we determined the fraction of the community formed by unique donor ASVs, that is, ASVs present in the donor's microbiota (i.e., FMT product) and patient's post-FMT microbiota but not the patient's pre-FMT microbiota. For patients who had both T2 and T3 samples, the larger of the two fractions was used. This fraction, denoted as dm_f , was used as a surrogate for donor microbiota engraftment, similar to previous studies (17, 20, 21). Continuous covariates included patient's age, interval between FMT and the last day of antibacterial antibiotics, interval between FMT and HCT or day 1 of induction chemotherapy, Shannon diversity of donor microbiota, Shannon diversity of patient's T1 microbiota, relative abundance of the 31 most abundant genera among T1 microbiota (patient samples), and relative abundance of the 31 most abundant genera in donor microbiota. Given that the number of covariates greatly exceeded the sample size, and some multicollinearity was expected, dm_f was modeled using 5-fold cross-validated least absolute shrinkage and selection operator (LASSO)-regularized linear regression, as implemented in the R package *glmnet* (v.4.1.7; ref. 24).

Regression coefficients were chosen at one SE from the cross-validated minimum value of lambda (tuning factor) and averaged across validation folds. By forcing the sum of the absolute value of the regression coefficients to be less than a fixed value, LASSO forces most coefficients to zero, leaving in only variables that are most likely true correlates of the outcome. Because categorical variables are not suitable as predictors in LASSO regression, we analyzed their association with dm_f separately using a univariate Wilcoxon test. Covariates in this analysis included: cohort (HCT vs. AML), stool donor, conditioning/chemotherapy intensity, and GVHD prophylaxis (PTCy-based vs. not). For GVHD prophylaxis, the analysis was limited to the HCT cohort. A Wilcoxon test was used to estimate the *P* values. LASSO variables that remained in the final model and statistically significant categorical variables from univariate analysis were ultimately included in a conventional linear regression model for dm_f . Regression coefficients and their 95% confidence intervals (CI) were determined.

In the second stage, we evaluated whether the extent of microbiota replacement by unique donor taxa is associated with the probability of grade II–IV aGVHD. This analysis included patients with dm_f in the FMT arm of the HCT cohort (27 patients). To this end, we compared dm_f between patients who subsequently developed grade II–IV aGVHD and those who developed grade I or no aGVHD. A Wilcoxon test was used to estimate the *P* value.

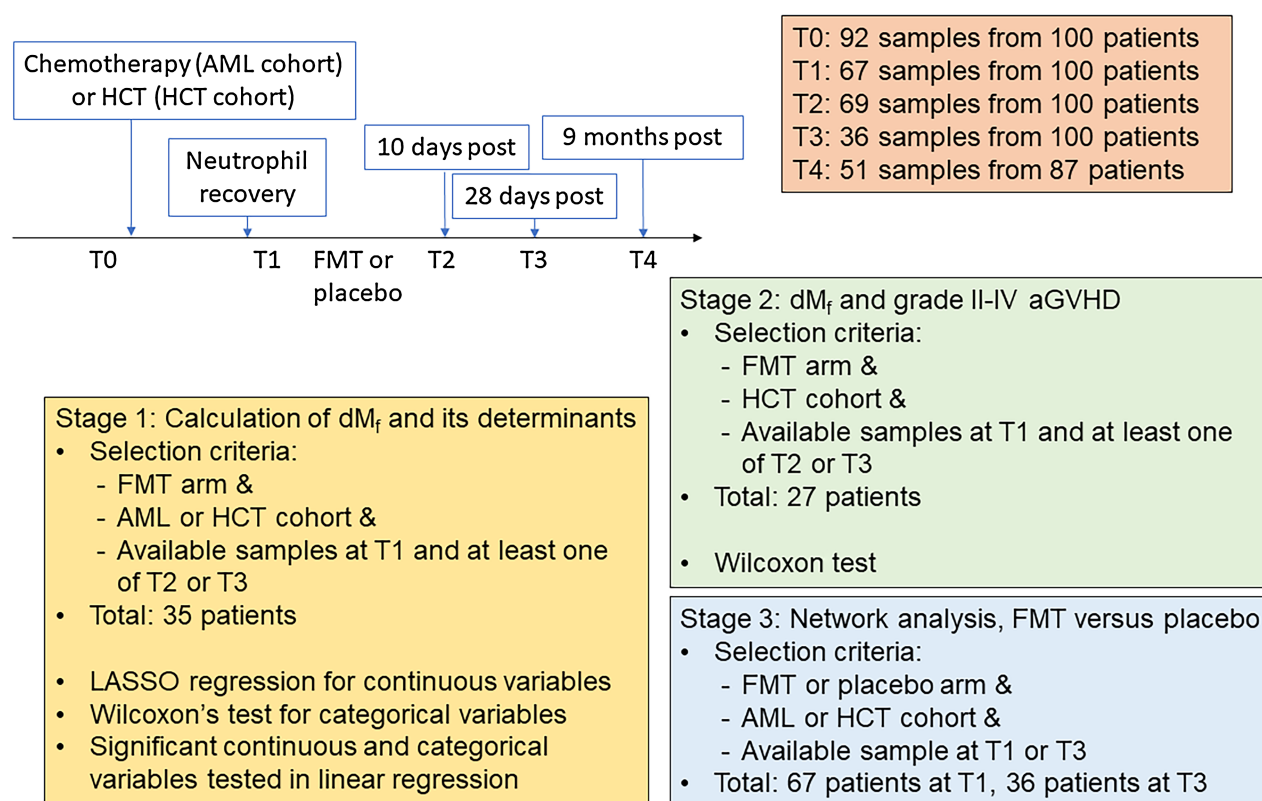


Figure 1.

Study schema. Schedule of sample collection, the number of samples and patients at each timepoint, and the three main stages of analysis are shown. Each box in the bottom (yellow, blue, or green) shows the subset of the entire cohort that was relevant for the corresponding analysis. The yellow box, for example, is the analysis of engraftment. It lists the criteria used to select relevant samples from relevant subjects, leading to samples from 35 patients. The blue box indicates a different analysis, requiring a different (though overlapping) subset of samples and patients. Similarly, the green box is a different analysis with different (though overlapping) subsets of patients and samples. The nature of each analysis dictated the subsets of samples and patients to be used. The orange box at the top right lists the total pool from which these samples and patients were selected. Each patient and sample in this pool contributed to at least one of the objectives.

In the third stage, we compared network properties of the microbiome between FMT and placebo arms at T1 and T3 timepoints in both cohorts. Network analysis was performed in the R package *NetCoMi* (25) and with ASVs aggregated at the genus level. To construct the networks, we used SparCC (26) to estimate associations between taxa and dissimilarity between samples. SparCC uses a variation of the pseudocount method to handle microbiome data sparsity and zero abundances, and explicitly considers microbiome data compositionality when estimating correlation values. The unsigned transformation was used to transform the estimated associations into dissimilarities resulting in networks in which edge weight between strongly correlated taxa is high, irrespective of the sign. To select edges present in the network, the default threshold of 0.3 was used as the minimum absolute association between the two genera (nodes). Centrality measures included degree, betweenness, and closeness. The degree of a node is defined as the number of its adjacent nodes. The betweenness centrality of a node is the number of shortest paths between any two nodes in the graph passing through that node. Betweenness centrality is a proxy for the node's location in relation to all other nodes, with higher/lower values associated with nodes located in the core/periphery of the network. The closeness centrality of a node is the average distance of this node to any other node. Clusters were identified using greedy modularity optimization. To compare networks between the groups (FMT vs. placebo), we calculated the Jaccard index (27). The Jaccard index (ranging from 0 to 1) quantifies, for each centrality measure, how similar the sets of the most central nodes are between the two networks. A value of 1 for this index means the sets of the most central nodes are exactly equal in both networks, while a value of 0 indicates that the most central nodes are completely different. The default centrality threshold of 0.75 was used to identify the most central nodes. Two-hundred permutations were used to test centrality measures and global network properties for group differences. For the Jaccard index, the *P* value indicates the probability of the index from a random permutation of the most central nodes between group 1, group 2, and the intersection of groups 1 and 2 being smaller than the calculated index. Thus, a significant *P* value means the Jaccard's index is likely zero, that is, the sets of the most central nodes in the two groups have little similarity. Permutation *P* values for centrality measures of different genera were adjusted using the local FDR method (28).

Preprocessing of sequencing data included filtering singletons, ASVs not present at a relative abundance of at least 0.1% in more than one sample, and samples with fewer than 5,000 reads. R 4.2.0 was used for all analyses and, unless stated otherwise, a two-sided *P* value of <0.05 was considered statistically significant.

Data availability

Raw sequence reads were uploaded to the NCBI Sequence Read Archive and are accessible under BioProject ID SRP287850. The custom R code is available from the corresponding author by email request. The trial protocol is available as a Supplementary Material.

Results

Cross-validated LASSO-regularized regression identifies pre-FMT microbiota diversity as the only determinant of donor microbiota engraftment

Supplementary Table S1, published previously (14), summarizes baseline characteristics of the two arms in each cohort. Supplementary Table S2 shows the representativeness of study participants. Donor microbiota engraftment was defined as the fraction of post-FMT

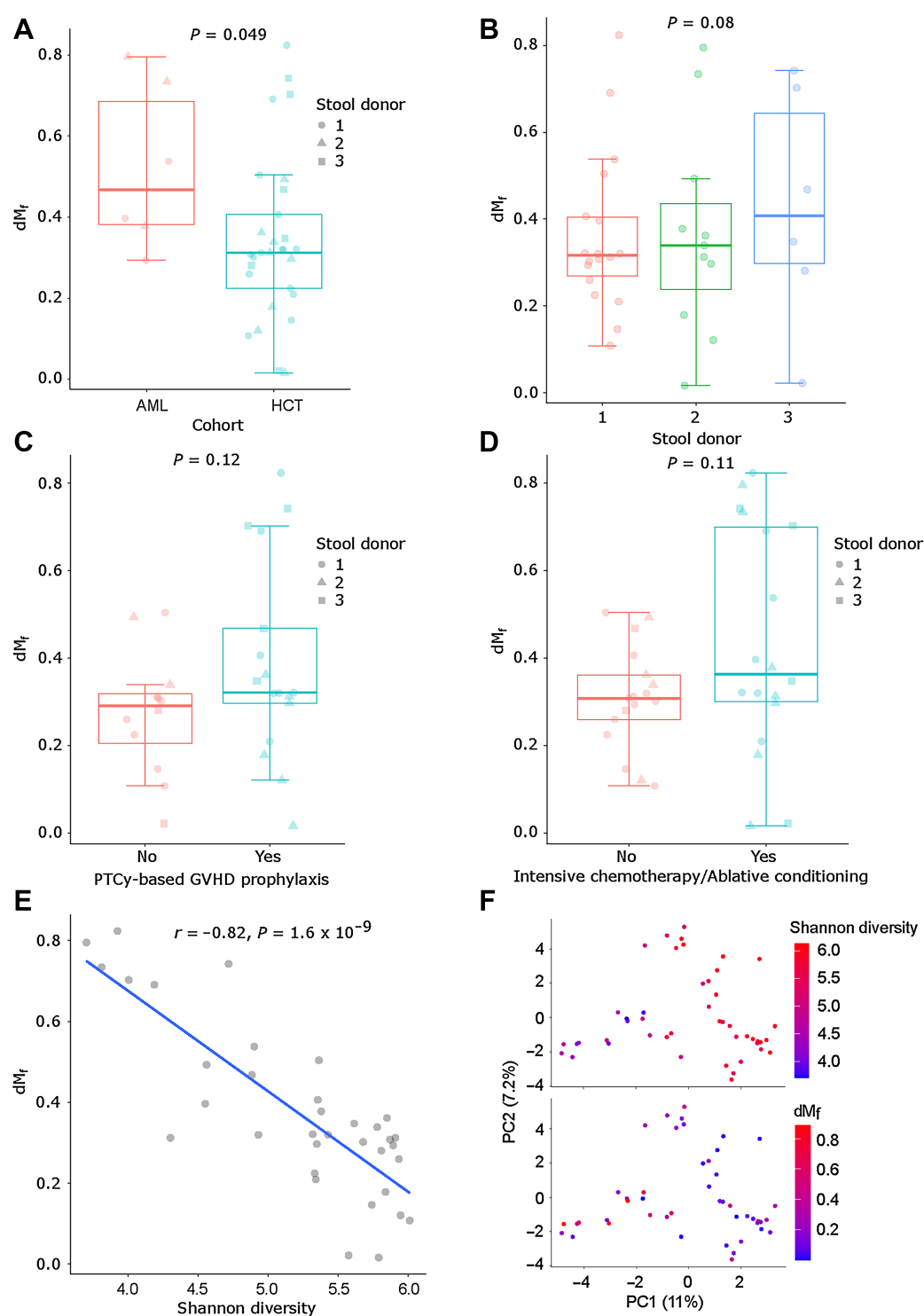
microbiota formed by unique donor ASVs (dM_f). This index could be calculated for 35 patients. Median dM_f at day 10–28 post-FMT was 32.0% (range, 1.6%–82.4%). In univariate analysis for categorical variables, dM_f was higher in the AML cohort (median 46.7%; range, 29.4%–79.5%) than the HCT cohort (median 31.2%; range, 1.6%–82.4%; *P* = 0.049; Fig. 2A). In contrast, no association was found between dM_f and FMT donor, GVHD prophylactic regimen, or conditioning/chemotherapy intensity (Fig. 2B–D). In LASSO regression with the remaining continuous covariates, only pre-FMT microbiota diversity was associated (inversely) with dM_f (Supplementary Fig. S1; Supplementary Data S1). Finally, we included cohort and pre-FMT diversity in a multivariable linear regression with dM_f as the outcome variable. In this analysis (adjusted R^2 , 0.66), only pre-FMT alpha diversity remained a significant predictor of dM_f (regression coefficient for diversity -0.24 ; 95% CI, -0.31 to -0.18 ; *P* = 1.5×10^{-8} ; regression coefficient for HCT vs. AML cohort -0.03 ; 95% CI, -0.15 to 0.09 ; *P* = 0.61). dM_f was strongly and inversely correlated with pre-FMT microbiota diversity (Pearson correlation coefficient -0.82 , *P* = 1.6×10^{-9} ; Fig. 2E). A similar pattern is seen in ordination plots in Fig. 2F.

Greater donor microbiota engraftment is associated with less grade II–IV aGVHD

A total of 12 grade II–IV aGVHD events occurred in FMT recipients of the HCT cohort for whom engraftment could be calculated. FMT product from 3 donors was used for patients included in this analysis (donor 1: 14 patients, donor 2: 7 patients, donor 3: 6 patients). Among FMT recipients from these donors, 6 (42.9%), 3 (42.9%), and 3 (50%) patients developed grade II–IV aGVHD, respectively. dM_f was significantly higher among patients who subsequently developed grade I or no aGVHD than those who developed grade II–IV aGVHD (median 33.9%; range, 1.6%–74.3% vs. median 25.3%; range, 2.2%–34.8%; *P* = 0.006; Fig. 3A). We confirmed this finding in time-to-event cumulative incidence analysis using Cox proportional hazards regression. The potential predictors in this model were dM_f (less vs. greater than median among all samples) and GVHD prophylaxis (PTCy-based vs. other) and the outcome variable was grade II–IV aGVHD. Competing risk (death without GVHD) occurred only in 1 patient; thus competing risk analysis was not included for simplicity. The cumulative incidence of grade II–IV aGVHD by day 180 was significantly lower in the group with greater-than-median engraftment than the group with less-than-median engraftment [14.3% (95% CI, 2.1–37.5) vs. 76.9% (95% CI, 39.7–92.8); Fig. 3B], with a HR of 0.12 (95% CI, 0.02–0.57; *P* = 0.008 from a Gray test).

Next, we tried to identify candidate taxa that might mediate the apparent protective effect of FMT against aGVHD. We focused on patients in the FMT arm without grade II–IV aGVHD and T1/T3 sample pairs from the same patients (12 samples from 6 patients), and searched for differentially abundant genera between the two timepoints. Genera analyzed included those with a minimum relative abundance of 0.5% in at least 25% of the samples. Despite using a relatively relaxed significance threshold of 0.10 for corrected *P* values, none of the 24 genera approached statistical significance (Supplementary Fig. S2). This could be attributed to our small sample size. Alternatively, species and strains not distinguishable at the genus level may be responsible for FMT effects. Short amplicon sequencing does not provide sufficiently high resolution for these levels of taxonomy.

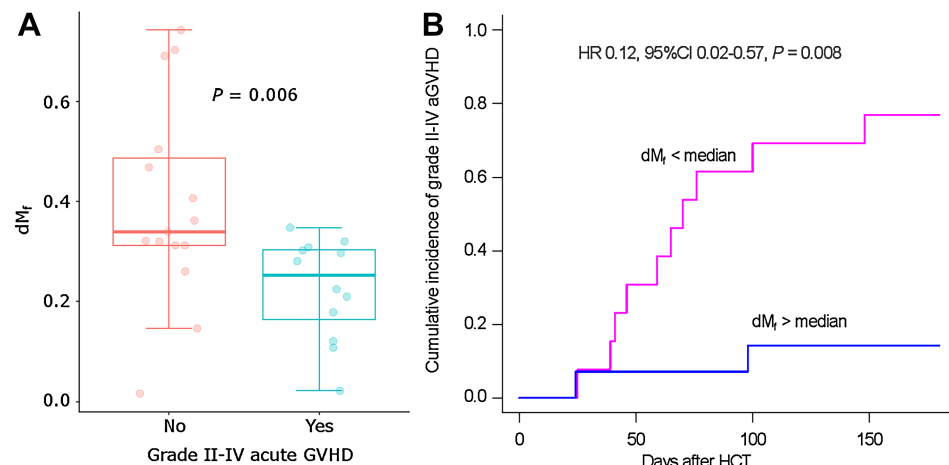
In our previous report and those by other groups (14, 29), FMT improved microbiota alpha diversity. Given the negative association between alpha diversity and subsequent aGVHD in observational studies (6, 8), we evaluated whether FMT weakens this association.

**Figure 2.**

Univariate analysis of dMf. **A**, Comparison between the two cohorts for dMf. **B**, Comparison between stool donors for dMf. Donor 4 is not included because none of the patients receiving product from this donor had the T1 and at least one of T2 or T3 stool samples for dMf determination. **C**, Comparison between different GVHD prophylactic regimens for dMf. **D**, Comparison between different intensities of induction chemotherapy (AML cohort) or conditioning regimen (HCT cohort) for dMf. **E**, Pearson correlation between dMf and pre-FMT (T1) alpha diversity as measured by the Shannon index. As alpha diversity increases along the x-axis, donor microbiota engraftment decreases along the y-axis. **F**, Ordination plots using Aitchison distance with color coding for Shannon diversity (top) and dMf (bottom). Each point is a sample, and its location is based on its overall microbiome composition. Only the HCT cohort was used to generate the plots in **C**. Both AML and HCT cohorts were used to generate the plots in the other panels. Stool donors in **A**, **C**, and **D** are shown in different shapes. The horizontal line within each box in **A–D** shows the median. A Wilcoxon test was used in **A–D** and a Pearson correlation test in **E**.

Figure 3.

Relationship between dM_f and grade II–IV aGVHD. **A**, Comparison between HCT recipients developing grade II–IV versus grade I or no aGVHD for dM_f . **B**, Time-to-event cumulative incidence analysis of grade II–IV aGVHD in groups with dM_f greater versus less than median among all samples. Only the HCT cohort was used to generate the plots. Two patients developed grade II–IV aGVHD before receiving FMT; these patients were not included in the analysis. The horizontal line within each box in **A** shows the median. A Wilcoxon test was used in **A** and a Gray test in **B**.



To this end, we built a Cox proportional hazards model with T1 Shannon diversity and GVHD prophylaxis (PTCy-based vs. other) as potential predictors for grade II–IV aGVHD in time-to-event, cumulative incidence analysis. One model was built for the placebo arm and one for the FMT arm. The HR for diversity in the placebo arm was 0.4 ($P = 0.59$), in a direction consistent with previous reports but without statistical significance likely due to the small sample size (15 patients). In contrast, in the FMT arm, HR for diversity was 3.4 ($P = 0.06$), indicating a trend opposite to the placebo arm. This finding is consistent with our observed association between lower donor microbiota engraftment and more grade II–IV aGVHD. Specifically, higher diversity before FMT predicts worse engraftment and more grade II–IV aGVHD.

Network analysis identifies FMT effects on community properties of microbiota networks

Some aspects of injury to the microbiota may recover spontaneously, while others may require interventions such as FMT. We demonstrated *Blautia* recovery and alpha diversity restoration to be examples for these two scenarios, respectively (14). The randomized design of the trial allowed us to distinguish between the two possibilities. Here, we used the same concept to investigate microbiota network properties that are differentially affected by FMT.

At T1 (pre-FMT/placebo), no differences were apparent between FMT and placebo arms in any of the studied microbiota network

properties (Table 1). In contrast, the number of network components at T3 (28 days after FMT/placebo) was significantly larger after FMT than placebo (mean 3 vs. 1, 200 permutations, $P = 0.04$; Table 1). A component is a connected subnetwork that is not part of any larger connected subnetwork. This finding is consistent with the previously observed increase in alpha diversity after FMT (more than placebo; ref. 14), resulting in a larger number of taxa and connected components. In addition, all centrality measures at T3 indicated a significant difference in the sets of the most central nodes between the two arms ($P = 0.03$; Supplementary Table S3). Specifically, the three most central T3 genera by degree were *Lachnospirillum*, *Flavonifractor*, and *Bacteroides* (tied with *Bifidobacterium*) in the FMT arm versus *Paraprevotella*, *Clostridium sensu stricto* 2, and *Parabacteroides* (tied with *Blautia*) in the placebo arm. These groups had only three of their 10 most central genera in common (Fig. 4A). Using closeness as the measure of centrality, the three most central genera in the FMT arm were *Bacteroides*, *Lachnospirillum*, and *Blautia* (tied with *Flavonifractor*), as compared with *Paraprevotella*, *Blautia*, and *Parabacteroides* in the placebo arm. These groups had only two of their 10 most central genera in common (Fig. 4B). Using betweenness as the measure of centrality, the three most central genera in the FMT arm were *Blautia*, *Bacteroides*, and *Flavonifractor* (tied with UCG-002), as compared with *Blautia*, *Paraprevotella*, and *Clostridium innocuum* group in the placebo arm. These groups had only three of their 10 most central genera in common (Fig. 4C). Interestingly,

Table 1. Global network properties in FMT versus placebo arms at pretreatment and day 28 posttreatment timepoints.

Measure	Pretreatment (T1)			Day 28 posttreatment (T3)		
	FMT	Placebo	P	FMT	Placebo	P
Number of components	9	1	0.82	3	1	0.04
Clustering coefficient	0.46	0.27	0.19	0.34	0.46	0.42
Modularity	0.46	0.45	0.94	0.31	0.21	0.47
Positive edge percentage	80.4	57.4	0.21	48.7	56.6	0.22
Edge density	0.08	0.11	0.90	0.16	0.28	0.39
Natural connectivity	0.03	0.03	0.77	0.03	0.03	0.72

Note: P values are one sided and test the null hypothesis that the absolute value of difference between the two arms is zero. A component is a connected subnetwork that is not part of any larger connected subnetwork. The clustering coefficient measures the degree to which nodes in a network tend to cluster together. Modularity measures the strength of division of a network into modules (e.g., clusters). Networks with high modularity have dense connections between nodes within modules but sparse connections between nodes in different modules. Edge density is the fraction of edges present over all possible edges. Natural connectivity is the average eigenvalue of the adjacency matrix and is a measure of the network's robustness.

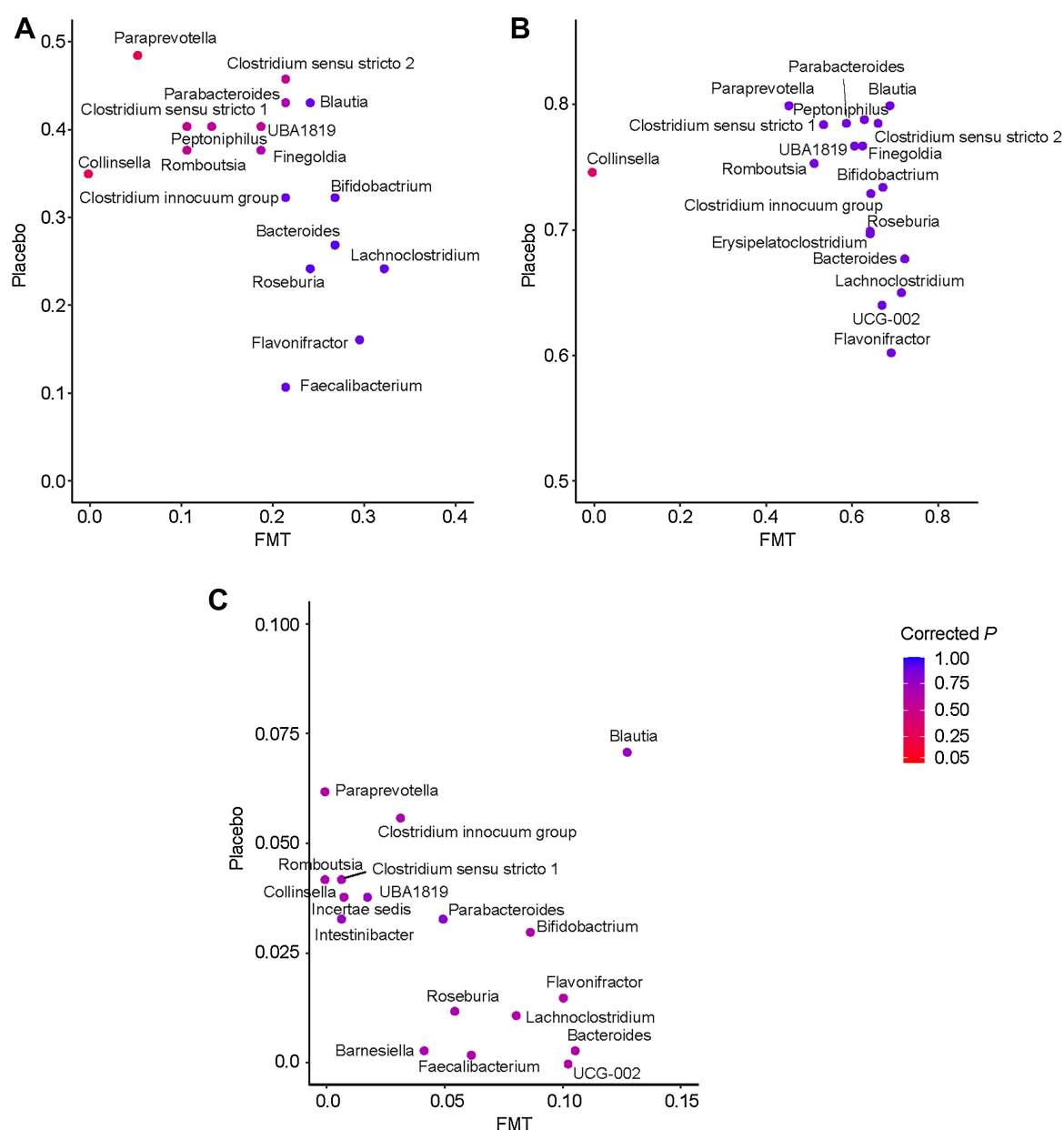


Figure 4.

Microbiota network analysis in T3 samples. **A**, Centrality of the 10 most central genera in each arm as measured by normalized degree. **B**, Centrality of the 10 most central genera in each arm as measured by normalized closeness. **C**, Centrality of the 10 most central genera in each arm as measured by normalized betweenness. *P* values comparing the two arms for each genus-level centrality measure are calculated from 200 permutations, corrected for multiple testing by the local FDR method, and visualized by a color gradient. None of the *P* values were statistically significant. In each plot, the centrality value of each genus is shown along the x-axis for the FMT arm and along the y-axis for the placebo arm. For example, *Paraprevotella* in **A** had greater centrality by normalized degree in the placebo arm than in the FMT arm, but the difference did not reach statistical significance. In contrast, *Bacteroides* in **C** had greater centrality by normalized betweenness in the FMT arm compared with placebo, but the difference did not reach statistical significance. Each plot includes fewer than 20 datapoints because the set of the most central genera in FMT and placebo arms had members in common.

Blautia was one of the most central genera in both arms and by all three centrality measures. **Figure 5** visualizes T3 microbiota networks in both arms.

Finally, as keystone species tend to have a high degree centrality, high closeness centrality, and low betweenness centrality (30), we explored whether the most central genera by degree

and closeness in FMT and placebo arms were among those in the 10 least central genera by betweenness. We found that none of the former group of genera (*Lachnoclostridium*, *Flavonifractor*, *Bacteroides*, *Blautia*, and *Bifidobacterium*) were among the latter group. This finding argues against significant appearance of keystone species postintervention.

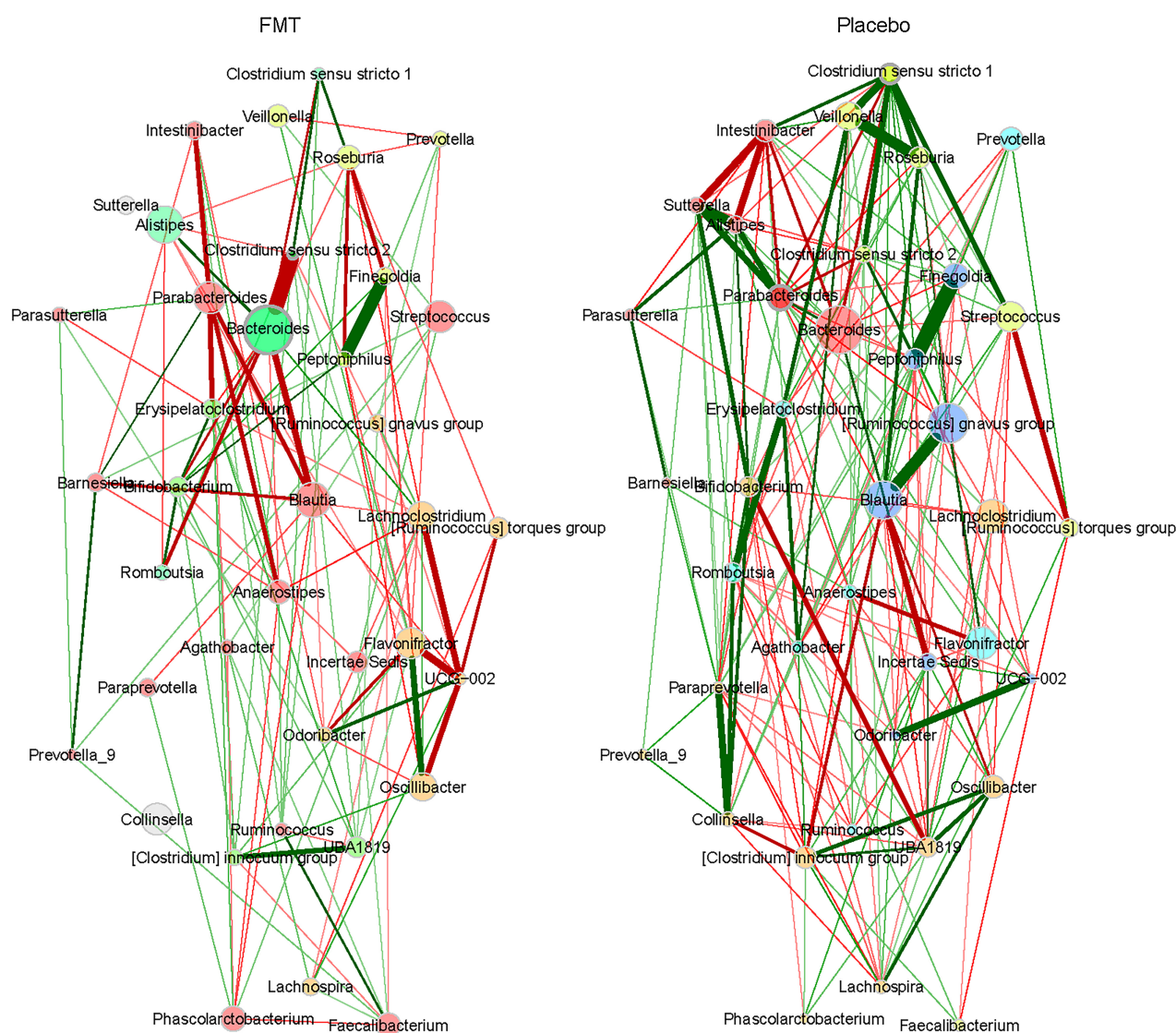


Figure 5.

T3 microbiota networks in FMT and placebo arms. Microbiota networks at the T3 timepoint in each arm are shown. Each genus is shown as a node. Nodes in the same cluster are colored the same. The size of each node is proportional to its centered log-ratio abundance. Green edges indicate positive associations while red edges represent negative associations. Edge weight indicates the strength of association. Nodes in the two networks are shown in the same position for ease of comparison. Nodes with no edges in either arm were removed.

Discussion

Correlative analysis of this randomized trial revealed three main findings. First, the inverse association between pre-FMT microbiota diversity and dM_f suggests that microbial communities with lower diversity are more receptive to novel taxa of donor origin and more amenable to modulation by FMT. This is consistent with the results of a large analysis of 14 FMT trials in five pathologies (recurrent *Clostridium difficile* infection, inflammatory bowel disease, metabolic syndrome, drug-resistant Enterobacteriaceae colonization, and immune checkpoint inhibitor-refractory melanoma; ref. 31). Second, we found that patients with subsequent grade II–IV aGVHD had a lower dM_f than those with grade I or no aGVHD. Collectively, these data provide evidence for potential efficacy of FMT in preventing

aGVHD, especially in patients with more severe dysbiosis. Third, our network analysis for the first time demonstrated major FMT effects on the wiring of microbial communities. FMT altered taxon centrality and conferred central locations within the microbiota network to genera such as *Lachnospira*, *Flavonifractor*, *Bacteroides*, *Blautia*, and *Bifidobacterium*. Three of these genera (*Lachnospira*, *Flavonifractor*, and *Blautia*) are known butyrate producers in the Clostridia class, while *Bifidobacterium* is a genus under the Actinobacteria phylum. *Bifidobacteria spp.* are among species with the highest engraftment rates after FMT in other settings (20). In our analysis, Actinobacteria had the highest rate of donor attribution among phyla (14). Notably, changes in microbiome network wiring were not accompanied by the emergence of keystone taxa. Keystone species are

typically infrequent taxa with low functional redundancy that regulate the functionality of the community as a whole and help hold the community together. On the basis of our findings, the effect of FMT on the microbiota does not seem to be via keystone species. The precise mechanisms by which the new wirings of the community lead to its differential function with potential clinical impact require further research.

FMT has been used with success in treating steroid-refractory aGVHD (17, 18). The current analysis supports potential efficacy of FMT also in the prophylactic setting. Importantly, the GVHD prophylactic regimen did not correlate with dM_f. Therefore, the observed association between dM_f and aGVHD was not confounded by the GVHD prophylactic regimen. Pre-FMT antibiotic conditioning has been associated with improved donor microbiota engraftment in patients with melanoma refractory to immune checkpoint blockade (32, 33). A brief course of rationally selected antibiotics before FMT could be tested in clinical trials as a strategy to improve microbiota engraftment.

Antibiotics are the main drivers of microbiota diversity loss in HCT recipients (34, 35) and specific antibiotic exposure patterns (types and timing) were associated with higher risk for aGVHD in a recent analysis (36). According to the results of the current analysis connecting pre-FMT microbiota diversity, microbiota modulation by FMT, and grade II–IV aGVHD, antibiotic exposures early after HCT may be used as an eligibility criterion in future prophylactic FMT trials. An alternative approach to identify patients with lower microbiota diversity is by measuring biomarkers of diversity such as urinary 3-indoxyl sulfate (37) and fecal chromogranin A (38).

The main limitation of the current analysis is related to the heterogeneity of variables that may influence the microbiota and aGVHD risk. When the trial was designed, and even at present, the ideal patient subsets for FMT were not known. Acknowledging that this trial would not be a definitive trial even for its primary endpoint for which it was powered, we designed the study as an “all comers” study, with few specific eligibility criteria. Demonstrating safety and feasibility of an intervention considered potentially high-risk at the time in a randomized trial were the main goals, hoping that the trial would provide novel information for future definitive trials. Nevertheless, the resulting heterogeneity in patient- and treatment-related characteristics limited the analysis. Both clinical and microbiota effects of FMT were likely diluted due to this heterogeneity, and thus underestimated.

To date, posttransplant GVHD prophylaxis has been based on the use of immunosuppressive medications with various toxicities and suboptimal efficacy. We have provided evidence for the potential of FMT as a novel nonimmunosuppressive adjunct to the current GVHD-prophylactic standards of care. A randomized trial powered for aGVHD as the primary endpoint and with subjects selected from those expected or documented to have more severe microbiota injury and lower microbiota diversity is warranted. Harnessing the

microbiota as a prophylactic measure against GVHD would represent a transformative conceptual advancement and a new direction in GVHD prophylaxis, potentially leading to improved quality of life and less mortality after transplantation.

Authors' Disclosures

A. Rashidi reports grants from NIH, NCI, University of Minnesota, and Achieving Cures Together during the conduct of the study, as well as consulting fees from Seres Therapeutics, Ltd and Emmes DSMB outside the submitted work. X. Luo reports grants from NIH and National Center for Advancing Translational Sciences during the conduct of the study. S.G. Holtan reports other support from Incyte, Vitrac Therapeutics, Ossium, CSL Behring, and Sanofi outside the submitted work. D.J. Weisdorf reports grants from Incyte outside the submitted work. A. Khoruts reports a patent for U.S. Patent No. 9,968,638 issued to Finch Therapeutics and a patent for U.S. Patent No. 10,028,980 issued to Finch Therapeutics. No disclosures were reported by the other authors.

Authors' Contributions

A. Rashidi: Conceptualization, data curation, formal analysis, supervision, funding acquisition, investigation, visualization, methodology, writing—original draft, project administration. **M. Ebadi:** Data curation, writing—review and editing. **T.U. Rehman:** Data curation. **H. Elhusseini:** Data curation. **D. Kazadi:** Data curation, writing—review and editing. **H. Halaweish:** Investigation. **M.H. Khan:** Investigation. **A. Hoeschen:** Project administration. **Q. Cao:** Formal analysis. **X. Luo:** Formal analysis. **A.J. Kabage:** Resources, project administration. **S. Lopez:** Resources, project administration. **S.G. Holtan:** Conceptualization, funding acquisition, project administration, writing—review and editing. **D.J. Weisdorf:** Writing—review and editing. **A. Khoruts:** Conceptualization, funding acquisition, writing—review and editing. **C. Staley:** Conceptualization, data curation, formal analysis, investigation, methodology, writing—review and editing.

Acknowledgments

This work is supported by NIH grant KL2TR002492, NIH grant UL1TR002494, NCI grant P30CA07759, University of Minnesota Chainbreaker grant, and Funds from Achieving Cures Together.

The content is solely the responsibility of the authors and does not represent the official views of the listed organizations. The funders had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and final responsibility for the decision to submit for publication.

DNA sequencing of the fecal samples was performed by the University of Minnesota Genomics Center. Sequence data were analyzed using the resources of the Minnesota Supercomputing Institute.

The publication costs of this article were defrayed in part by the payment of publication fees. Therefore, and solely to indicate this fact, this article is hereby marked “advertisement” in accordance with 18 USC section 1734.

Note

Supplementary data for this article are available at Clinical Cancer Research Online (<http://clincancerres.aacrjournals.org/>).

Received August 4, 2023; revised September 18, 2023; accepted September 29, 2023; published first October 3, 2023.

References

- Shono Y, van den Brink MRM. Gut microbiota injury in allogeneic haematopoietic stem cell transplantation. *Nat Rev Cancer* 2018;18:283–95.
- Rashidi A, Weisdorf DJ. Microbiota-based approaches to mitigate infectious complications of intensive chemotherapy in patients with acute leukemia. *Transl Res* 2020;220:167–81.
- Peled JU, Gomes ALC, Devlin SM, Littmann ER, Taur Y, Sung AD, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med* 2020;382:822–34.
- Ilett EE, Jørgensen M, Noguera-Julian M, Nørgaard JC, Dagaard G, Helleberg M, et al. Associations of the gut microbiome and clinical factors with acute GVHD in allogeneic HSCT recipients. *Blood Adv* 2020;4:5797–809.
- Jenq RR, Taur Y, Devlin SM, Ponce DM, Goldberg JD, Ahr KF, et al. Intestinal blautia is associated with reduced death from graft-versus-host disease. *Biol Blood Marrow Transplant* 2015;21:1373–83.
- Burgos da Silva M, Ponce DM, Dai A, Devlin SM, Gomes AL, Moore GF, et al. Preservation of the fecal microbiome is associated with reduced severity of graft-versus-host disease. *Blood* 2022;140:2385–97.

7. Meedt E, Hiergeist A, Gessner A, Dettmer K, Liebisch G, Ghimire S, et al. Prolonged suppression of butyrate-producing bacteria is associated with acute gastrointestinal graft-vs-host disease and transplantation-related mortality after allogeneic stem cell transplantation. *Clin Infect Dis* 2022; 74:614–21.
8. Golob JL, Pergam SA, Srinivasan S, Fiedler TL, Liu C, Garcia K, et al. Stool microbiota at neutrophil recovery is predictive for severe acute graft vs host disease after hematopoietic cell transplantation. *Clin Infect Dis* 2017;65: 1984–91.
9. Stein-Thoeringer CK, Nichols KB, Lazrak A, Docampo MD, Slingerland AE, Slingerland JB, et al. Lactose drives enterococcus expansion to promote graft-versus-host disease. *Science* 2019;366:1143–9.
10. Hayase E, Hayase T, Jamal MA, Miyama T, Chang C-C, Ortega MR, et al. Mucus-degrading bacteroides link carbapenems to aggravated graft-versus-host disease. *Cell* 2022;185:3705–19.
11. Legoff J, Resche-Rigon M, Bouquet J, Robin M, Naccache SN, Mercier-Delarue S, et al. The eukaryotic gut virome in hematopoietic stem cell transplantation: new clues in enteric graft-versus-host disease. *Nat Med* 2017;23:1080–5.
12. Swimm A, Giver CR, DeFilipp Z, Rangaraju S, Sharma A, Antonova AU, et al. Indoles derived from intestinal microbiota act via type I interferon signaling to limit graft-versus-host disease. *Blood* 2018;132:2506–19.
13. Saqr A, Carlson B, Staley C, Rashidi A, Al-Kofahi M, Kaiser T, et al. Reduced enterohepatic recirculation of mycophenolate and lower blood concentrations are associated with the stool bacterial microbiome after hematopoietic cell transplantation. *Transplant Cell Ther* 2022;28:372.e1–9.
14. Rashidi A, Ebadi M, Rehman TU, Elhousseini H, Kazadi D, Halaweish H, et al. Randomized double-blind phase II trial of fecal microbiota transplantation versus placebo in allogeneic hematopoietic cell transplantation and AML. *J Clin Oncol* 2023 May 26 [Epub ahead of print].
15. El Jurdi N, Hoover A, O'Leary D, Cao Q, Gupta A, Ebens C, et al. Phase II study of myeloablative 7–8/8-matched allotransplantation with post-transplant cyclophosphamide, tacrolimus, and mycophenolate mofetil. *Transplant Cell Ther* 2023;29:576.e1–5.
16. Bolaños-Meade J, Hamadani M, Wu J, Al Malki MM, Martens MJ, Runaas L, et al. Post-transplantation cyclophosphamide-based graft-versus-host disease prophylaxis. *N Engl J Med* 2023;388:2338–48.
17. van Lier YF, Davids M, Haverkate NJE, de Groot PF, Donker ML, Meijer E, et al. Donor fecal microbiota transplantation ameliorates intestinal graft-versus-host disease in allogeneic hematopoietic cell transplant recipients. *Sci Transl Med* 2020;12:eaa28926.
18. Zhao Y, Li X, Zhou Y, Gao J, Jiao Y, Zhu B, et al. Safety and efficacy of fecal microbiota transplantation for grade IV steroid refractory GI-GvHD patients: interim results from FMT2017002 trial. *Front Immunol* 2021;12: 678476.
19. Kakihana K, Fujioka Y, Suda W, Najima Y, Kuwata G, Sasajima S, et al. Fecal microbiota transplantation for patients with steroid-resistant acute graft-versus-host disease of the gut. *Blood* 2016;128:2083–8.
20. Ianiro G, Punčochář M, Karcher N, Porcari S, Armanini F, Asnicar F, et al. Variability of strain engraftment and predictability of microbiome composition after fecal microbiota transplantation across different diseases. *Nat Med* 2022;28: 1913–23.
21. Porcari S, Benech N, Valles-Colomer M, Segata N, Gasbarrini A, Cammarota G, et al. Key determinants of success in fecal microbiota transplantation: from microbiome to clinic. *Cell Host Microbe* 2023;31:712–33.
22. Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJA, Holmes SP. DADA2: high-resolution sample inference from Illumina amplicon data. *Nat Methods* 2016;13:581–3.
23. Quast C, Pruesse E, Yilmaz P, Gerken J, Schweer T, Yarza P, et al. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Res* 2013;41:D590–6.
24. Friedman J, Hastie T, Tibshirani R. Regularization paths for generalized linear models via coordinate descent. *J Stat Softw* 2010;33:1–22.
25. Peschel S, Müller CL, Von Mutius E, Boulesteix A-L, Depner M. NetCoMi: network construction and comparison for microbiome data in R. *Brief Bioinform* 2021;22:bbaa290.
26. Friedman J, Alm EJ. Inferring correlation networks from genomic survey data. *PLoS Comput Biol* 2012;8:e1002687.
27. Real R, Vargas JM. The probabilistic basis of Jaccard's index of similarity. *Syst Biol* 1996;45:380–5.
28. Efron B. Local false discovery rates. Large-scale inference. Cambridge (UK): Cambridge University Press; 2010. p. 70–88.
29. Taur Y, Coyte K, Schluter J, Robilotti E, Figueroa C, Gjonbalaj M, et al. Reconstitution of the gut microbiota of antibiotic-treated patients by autologous fecal microbiota transplant. *Sci Transl Med* 2018;10:eaap9489.
30. Berry D, Widder S. Deciphering microbial interactions and detecting keystone species with co-occurrence networks. *Front Microbiol* 2014;5:219.
31. Podlesny D, Durdevic M, Paramsothy S, Kaakoush NO, Högenauer C, Gorkiewicz G, et al. Identification of clinical and ecological determinants of strain engraftment after fecal microbiota transplantation using metagenomics. *Cell Rep Med* 2022;3:100711.
32. Baruch EN, Youngster I, Ben-Betzalel G, Ortenberg R, Lahat A, Katz L, et al. Fecal microbiota transplant promotes response in immunotherapy-refractory melanoma patients. *Science* 2021;371:602–9.
33. Davar D, Dzutsev AK, McCulloch JA, Rodrigues RR, Chauvin J-M, Morrison RM, et al. Fecal microbiota transplant overcomes resistance to anti-PD-1 therapy in melanoma patients. *Science* 2021;371:595–602.
34. Willmann M, Vehreschild MJGT, Biehl LM, Vogel W, Dörfel D, Hamprecht A, et al. Distinct impact of antibiotics on the gut microbiome and resistome: a longitudinal multicenter cohort study. *BMC Biol* 2019;17:76.
35. Weber D, Hiergeist A, Weber M, Dettmer K, Wolff D, Hahn J, et al. Detrimental effect of broad-spectrum antibiotics on intestinal microbiome diversity in patients after allogeneic stem cell transplantation: lack of commensal sparing antibiotics. *Clin Infect Dis* 2019;68:1303–10.
36. Rashidi A, Gao F, Fredricks DN, Pergam SA, Mielcarek M, Milano F, et al. Analysis of antibiotic exposure and development of acute graft-vs-host disease following allogeneic hematopoietic cell transplantation. *JAMA Netw Open* 2023; 6:e2317188.
37. Weber D, Oefner PJ, Hiergeist A, Koestler J, Gessner A, Weber M, et al. Low urinary indoxyl sulfate levels early after transplantation reflect a disrupted microbiome and are associated with poor outcome. *Blood* 2015;126:1723–8.
38. Zhernakova A, Kurilshikov A, Bonder MJ, Tigchelaar EF, Schirmer M, Vatanen T, et al. Population-based metagenomics analysis reveals markers for gut microbiome composition and diversity. *Science* 2016;352:565–9.